Lecture 8 Scribe

L8_S2_A (1).pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 8 slides).Instructions:Use 'One-Shot Prompting'. I will provide you with an Example (The Shot) of how to convert a slide concept into a scribe format. You must follow this structure and style to generate the scribe for the Lecture 8 slides.EXAMPLE (THE SHOT):Input Context: 'Slide 1: Definition of Set. A set is a collection of distinct objects. Example: A = {1, 2, 3}. Notation: x ∈ A means x is in set A.'Output Scribe:'### 1. Fundamentals of SetsDefinition: A set is a well-defined collection of distinct objects.Notation:$A = \{1, 2, 3\}$ represents a set with elements 1, 2, and 3.$x \in A$ denotes that element x belongs to set A.'YOUR TASK:Now, generate the full lecture scribe for the attached Lecture 8 slides following the style and faithfulness of the example above.Faithfulness: Do not add outside information. Use exact definitions/notations from slides.Structure: Use clear headings (e.g., Sample Space, Events, Probability Axioms) as per the slides.Goal: Exam-ready reference material."

Got it — here is an **exam-ready Lecture 8 scribe**, written strictly from the provided slides and following the structure/style of your one-shot example. All definitions, formulas, and identities are taken directly from the Lecture 8 material

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8. Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables

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8.1 Gaussian Random Variable — Definition and Properties

Definition:

A Gaussian random variable is one whose PDF can be written in the general form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2}),$$

where:

- m is the mean,
- σ^2 is the variance.

Notation:

$$X \sim \mathrm{N}(m, \sigma^2).$$

8.2 Standard Forms for Gaussian Random Variables

Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt.$$

Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt.$$

Φ-Function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(-\frac{t^2}{2})\, dt.$$

Q-Function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp(-\frac{t^2}{2})\, dt.$$

Note:

$$Q(x) = 1 - \Phi(x).$$

8.3 Relation Between Φ-Function and Q-Function

Evaluating the CDF of a Gaussian RV

For $X \sim \mathrm{N}(m, \sigma^2)$,

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy.$$

With the change of variables $t = \frac{y-m}{\sigma}$,

$$F_X(x) = \Phi(\frac{x-m}{\sigma}).$$

Evaluating Tail Probabilities

$$\Pr(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy = Q(\frac{x-m}{\sigma}).$$

Key Identities

$$Q(x) = 1 - \Phi(x),$$

$$F_X(x) = 1 - Q(\frac{x-m}{\sigma}).$$

Remarks from slides:

- To evaluate the CDF of a Gaussian RV, evaluate the Q-function at the points.
- The Q-function is more natural for probabilities of the form $\Pr(X > x)$.

8.4 Example: Gaussian Probability Calculations

A random variable X has PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp(-\frac{(x+3)^2}{8}).$$

Comparing with the Gaussian PDF,

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2}),$$

we identify:

$$m = -3, \qquad \sigma^2 = 4, \qquad \sigma = 2.$$

Gaussian relations used:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right), \qquad \Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right).$$

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1) $\Pr(X \leqslant 0)$

$$\Pr(X \leqslant 0) = F_X(0) = \Phi\left(\frac{0-m}{\sigma}\right) = \Phi\left(\frac{3}{2}\right) = 1 - Q\left(\frac{3}{2}\right).$$

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2) $\Pr(X > 4)$

$$\Pr(X > 4) = Q\left(\frac{4-m}{\sigma}\right) = Q\left(\frac{7}{2}\right).$$

(Additional probabilities in the slide are handled using the same transformations into standard form and expressed using $\Phi(\cdot)$ or $Q(\cdot)$.)

8.5 Types of Continuous Random Variables (Outline)

The lecture also introduces the following continuous random variables (with examples discussed in class):

- Uniform Random Variable
- Exponential Random Variable
- Laplace Random Variable
- Gamma Random Variable (including graph and special cases)

(Problem solving, homework, and an in-class activity on Gaussian density estimation follow in the lecture.)

If you'd like, next we can turn this into a **formula sheet**, add **worked steps for the remaining example sub-parts**, or create **practice problems in the same style** for exam prep.

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Sure — below is the **full Lecture 8 exam scribe**, written strictly from the provided slides and formatted to mirror your one-shot example (compact headings + definition/notation blocks, no outside material added). All content is taken directly from the Lecture 8 slides

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8. Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables

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8.1 Gaussian Random Variable — Definition and Properties

Definition:

A Gaussian random variable is one whose PDF can be written in the general form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2}).$$

Notation:

$$X \sim \mathrm{N}(m, \sigma^2),$$

where m is the mean and σ^2 is the variance.

8.2 Expectation and Moments (from lecture notes)

Expected value of a function of RV X:

- Continuous RV:

$$E[g(X)] = \int g(x) f_X(x)\, dx.$$

- Discrete RV:

$$E[g(X)] = \sum_k g(x_k) P_X(x_k).$$

Linearity of expectation:

$$E[aX + b] = aE[X] + b.$$

For a sum:

$$E[\sum_{k=1}^{N} g_k(X)] = \sum_{k=1}^{N} E[g_k(X)].$$

Central Moments (as written on slides)

- Second central moment:

$$E[(X - \mu_X)^2] = \sigma_X^2.$$

- Third central moment (Skewness):

$$C_s = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}.$$

Interpretation (from slides): measures symmetry of the PDF about μ_X.

Positive value → right skewed.

Negative value → left skewed.

- Fourth central moment (Kurtosis):

$$C_k = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}.$$

Large value indicates a large peak near the mean.

8.3 Standard Forms for Gaussian Random Variables

Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt.$$

Complementary Error Function

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt.$$

Φ-Function (CDF of Standard Normal)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp(-\frac{t^2}{2})\, dt.$$

Q-Function (Gaussian Tail Function)

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp(-\frac{t^2}{2})\, dt.$$

Note:

$$Q(x) = 1 - \Phi(x).$$

8.4 Relation Between Φ-Function and Q-Function

Evaluating the CDF

For $X \sim \mathrm{N}(m, \sigma^2)$:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy.$$

With substitution $t = \frac{y-m}{\sigma}$:

$$F_X(x) = \Phi(\frac{x-m}{\sigma}).$$

Evaluating Tail Probabilities

$$\Pr(X > x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-m)^2}{2\sigma^2})\, dy = Q(\frac{x-m}{\sigma}).$$

Key Identities

$$Q(x) = 1 - \Phi(x),$$

$$F_X(x) = 1 - Q(\frac{x-m}{\sigma}).$$

Lecture remarks:

- To evaluate the CDF of a Gaussian RV, evaluate the Q-function at the points.
- The Q-function is more natural for probabilities of the form $\Pr(X > x)$.
- $\Phi(x)$ corresponds to area under the left tail; $Q(x)$ corresponds to area under the right tail.

8.5 Gaussian Example

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp(-\frac{(x+3)^2}{8}).$$

Comparing with:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-m)^2}{2\sigma^2}),$$

we identify:

$$m = -3, \qquad \sigma^2 = 4, \qquad \sigma = 2.$$

Gaussian relations:

$$F_X(x) = \Phi(\frac{x-m}{\sigma}), \qquad \Pr(X > x) = Q(\frac{x-m}{\sigma}).$$

Probability Results Shown in Slides

1. **Left-tail probability**

$$\Pr(X \leqslant 0) = F_X(0) = \Phi(\frac{0-m}{\sigma}) = \Phi(\frac{3}{2}) = 1 - Q(\frac{3}{2}).$$

2. **Right-tail probability**

$$\Pr(X > 4) = Q(\frac{4-m}{\sigma}) = Q(\frac{7}{2}).$$

(Other example parts in the slide are handled by converting to standard form and expressing answers using $\Phi(\cdot)$ and $Q(\cdot)$.)

8.6 Types of Continuous Random Variables (Lecture Outline)

The lecture further covers (with examples):

- Uniform Random Variable
- Exponential Random Variable
- Laplace Random Variable
- Gamma Random Variable
  - Graph and special cases
  - Example

Followed by:

- Homework problem
- Problem solving
- In-class activity: Gaussian density estimation

If you'd like, I can next condense this into a **one-page formula sheet** or generate **practice problems using only these slide relations** for exam revision.

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Gaussian relations:

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Probabilities Shown

1.

$$\Pr(X \leqslant 0) = F_X(0) = \Phi(\frac{0-m}{\sigma}) = \Phi(\frac{3}{2}) = 1 - Q(\frac{3}{2}).$$

2.

$$\Pr(X > 4) = Q(\frac{4-m}{\sigma}) = Q(\frac{7}{2}).$$

(Other parts are obtained by converting to standard form and expressing answers using $\Phi(\cdot)$ and $Q(\cdot)$.)

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